



AI Security

Real-Time Object Tracking and Obstruction
Detection System.

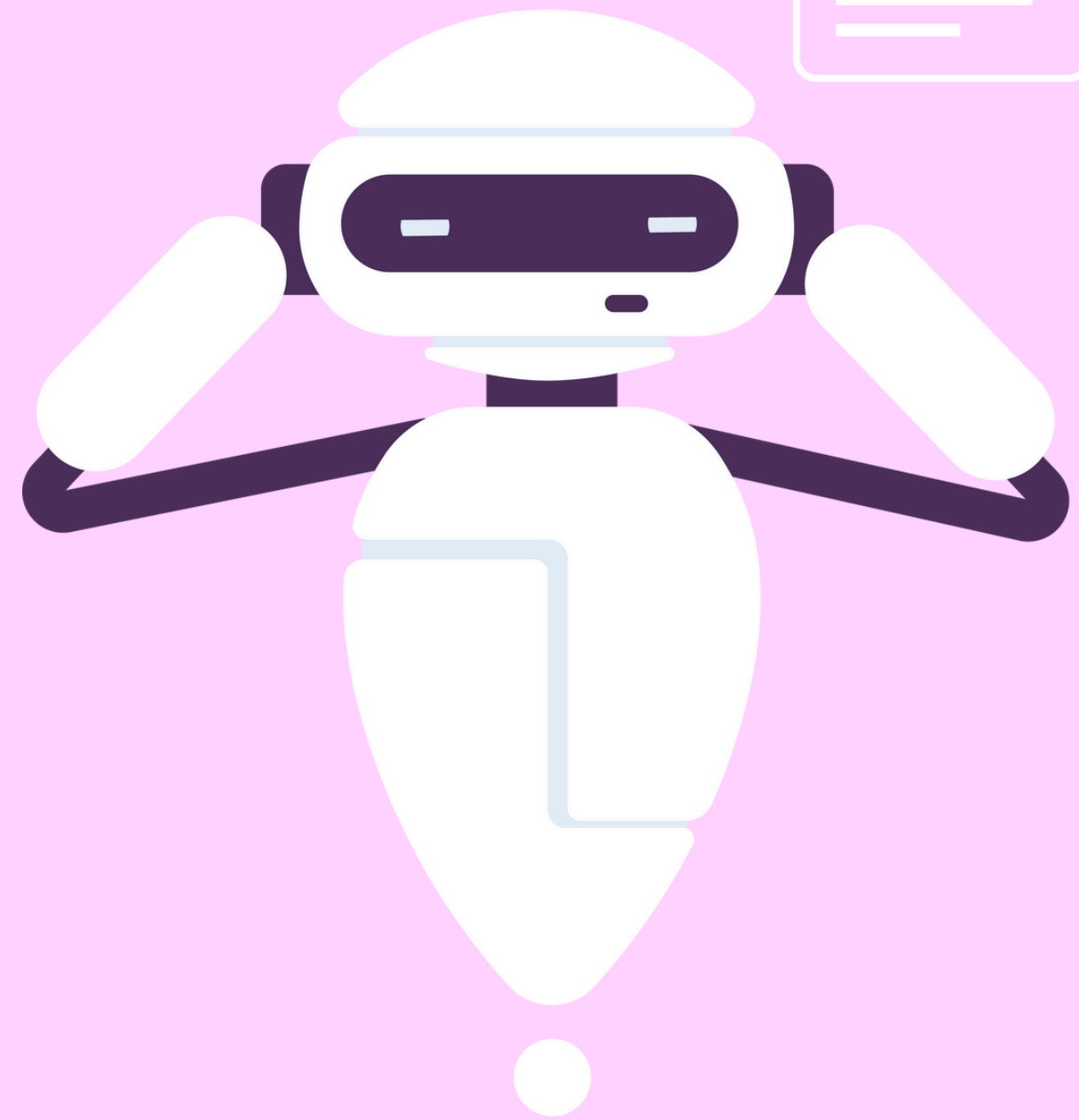
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problem statement

Develop an AI-powered security surveillance system that can detect and alert owners in real-time about potential theft or movement of their vehicles or valuable objects, even in low-light conditions or when obstructed.

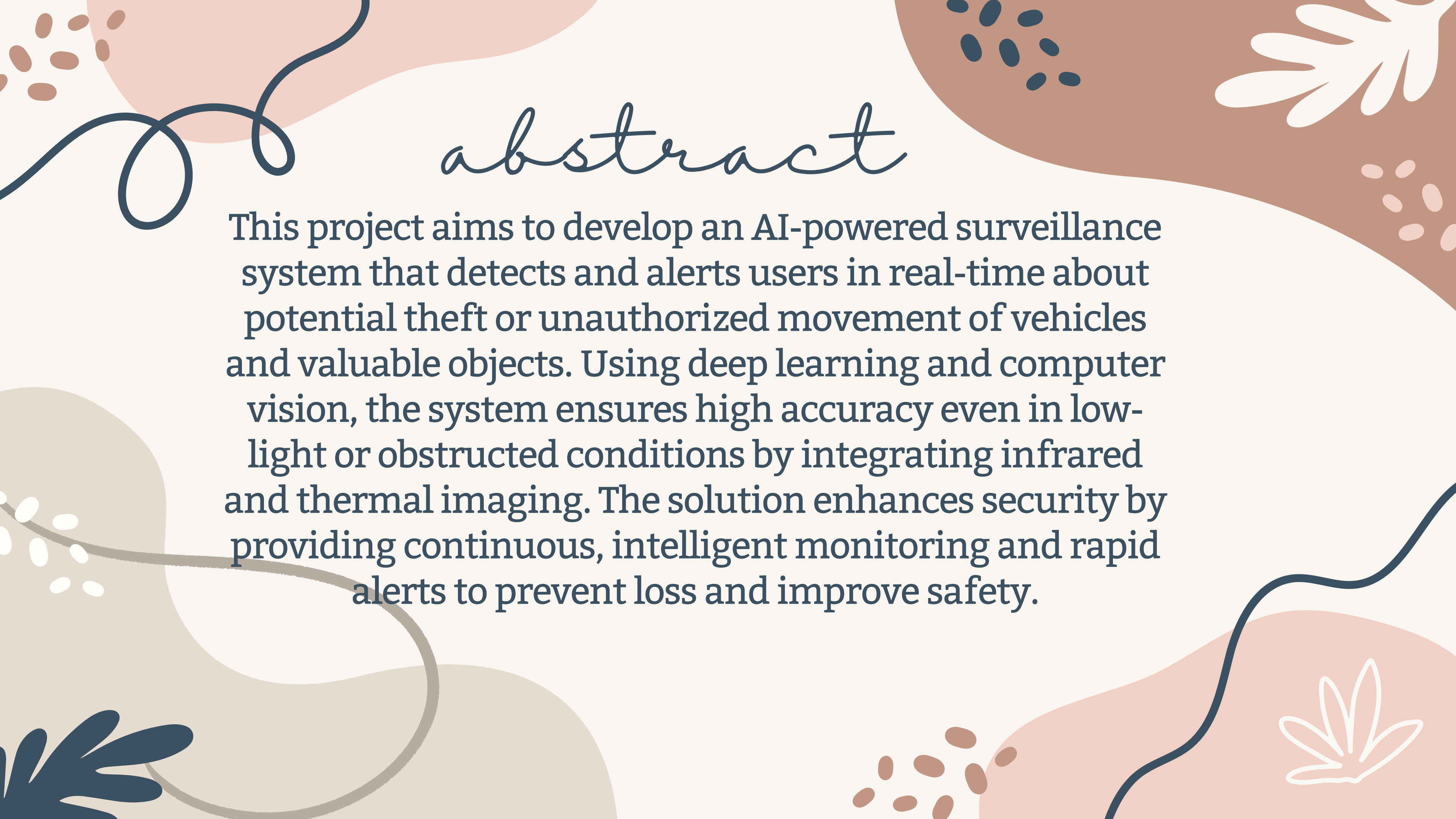


MOTIVATION

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With increasing incidents of theft and unauthorized access to vehicles and valuable assets, traditional surveillance systems often fall short due to poor performance in low-light or obstructed conditions. This project aims to develop an AI-powered security surveillance system that uses advanced computer vision and deep learning to detect and alert users in real-time about suspicious movement or potential theft. By enhancing accuracy, response time, and performance in challenging environments, the system provides a smarter, more reliable solution for modern security needs.





abstract

This project aims to develop an AI-powered surveillance system that detects and alerts users in real-time about potential theft or unauthorized movement of vehicles and valuable objects. Using deep learning and computer vision, the system ensures high accuracy even in low-light or obstructed conditions by integrating infrared and thermal imaging. The solution enhances security by providing continuous, intelligent monitoring and rapid alerts to prevent loss and improve safety.

Existing Solutions: A Comparison

Category	Narrow AI	General AI
Purpose	Narrow AI is programmed to perform only a limited range of predefined tasks in order to finish or solve a certain issue.	General AI will be self-aware and capable of carrying out any task that its “mind” can conjure up.
AI Model	Narrow AI employs preprogrammed fixed domain models.	The operating environment helps general AI reason and learn on its own.
Awareness	Self-awareness, artificial consciousness, and cognitive capabilities are lacking from narrow AI.	General AI will be regarded as being highly sophisticated, intelligent, and fully self-aware, which implies that it will have common sense, creativity, and the capacity for emotional expression.
Knowledge Transfer	Knowledge cannot be transferred from narrow AI to other areas or tasks.	General AI uses knowledge transfer to tackle new problems and domains.
AI Stages	Today’s AI	Future AI - Around 2040

Traditional Systems

CCTV cameras with manual monitoring. They are limited to recording and playback.

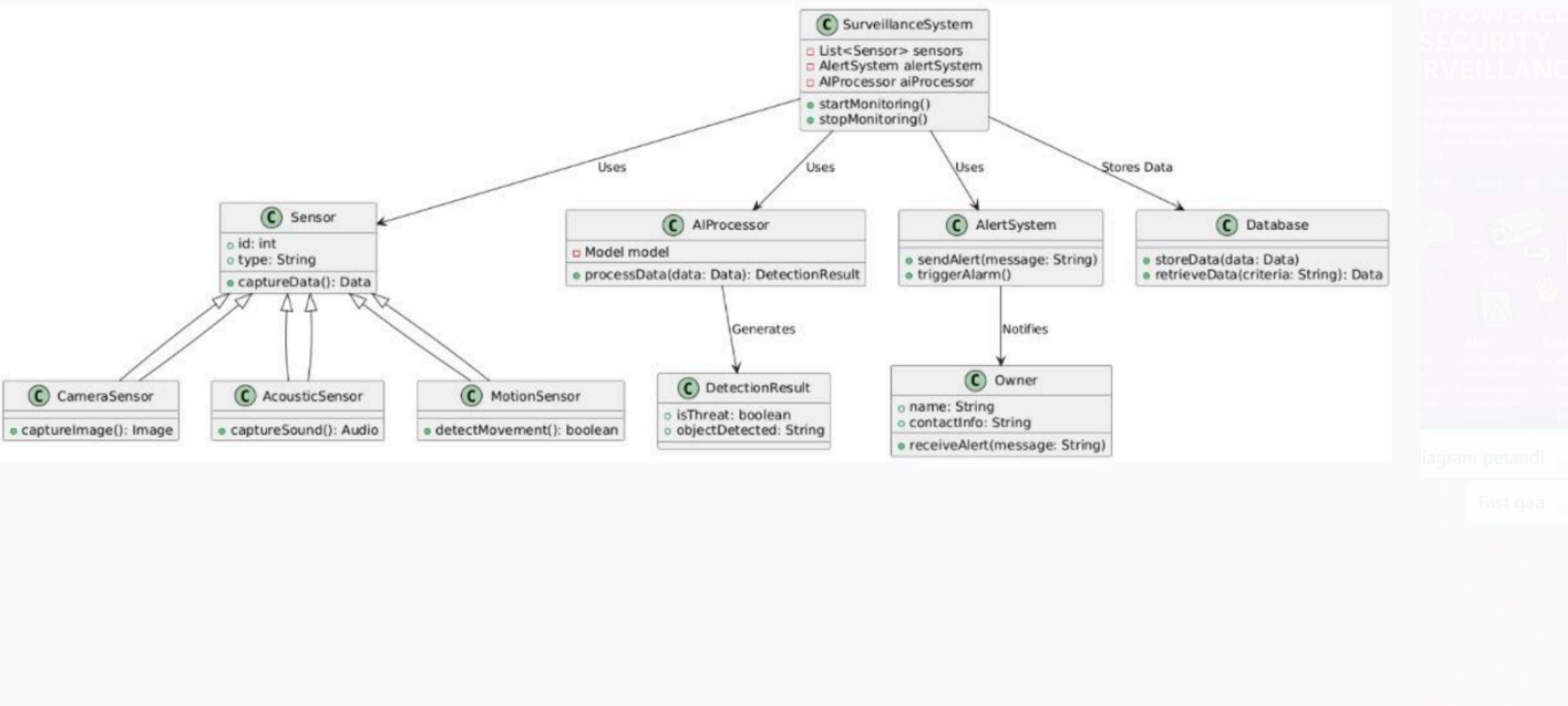
AI-Based Systems

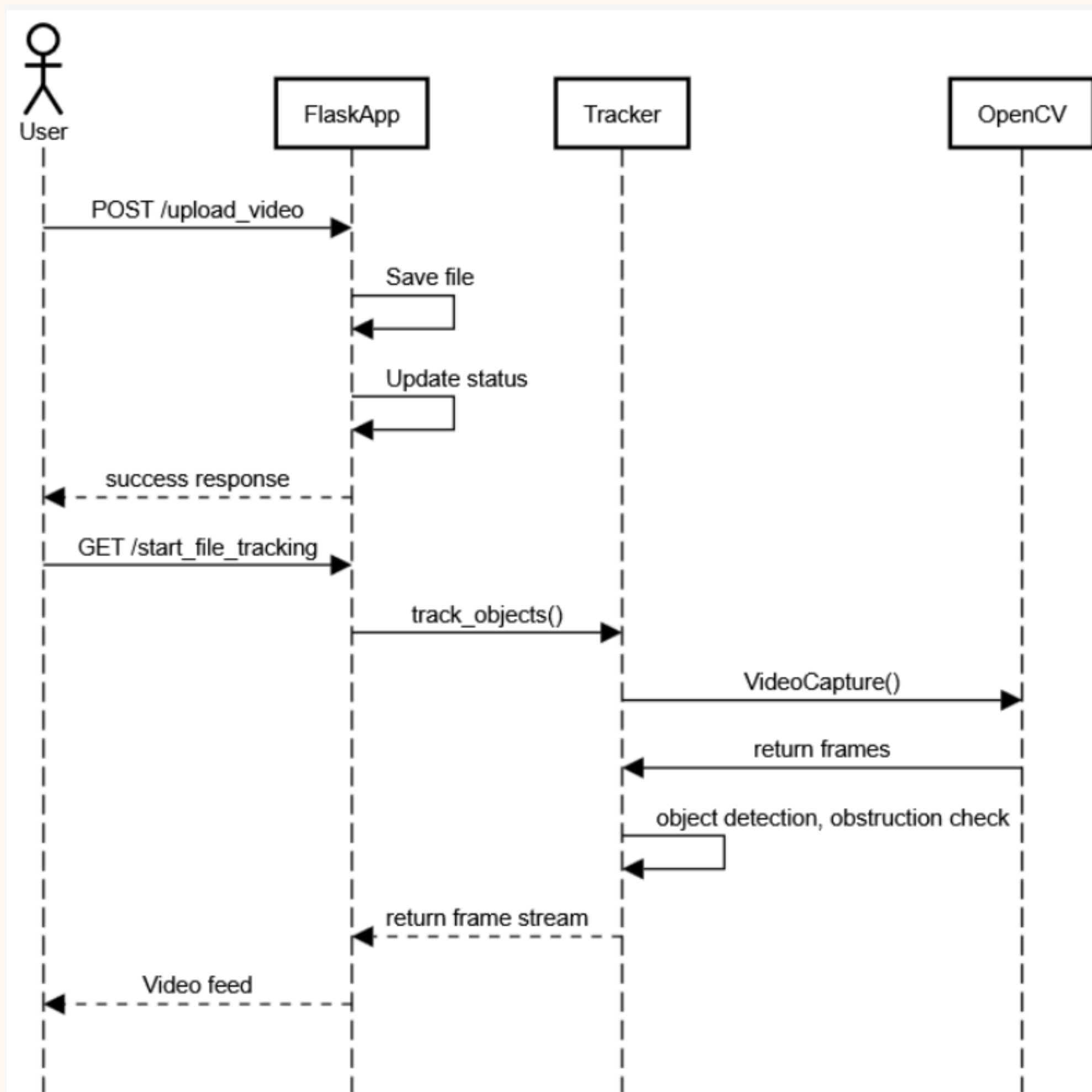
Object detection, tracking, and facial recognition. Also includes anomaly detection.

Limitations

High cost and limited customization. Lack of real-time obstruction detection.

CLASS DIAGRAM





Technology stack

1

Backend

Flask, OpenCV, Winsound.

2

Frontend

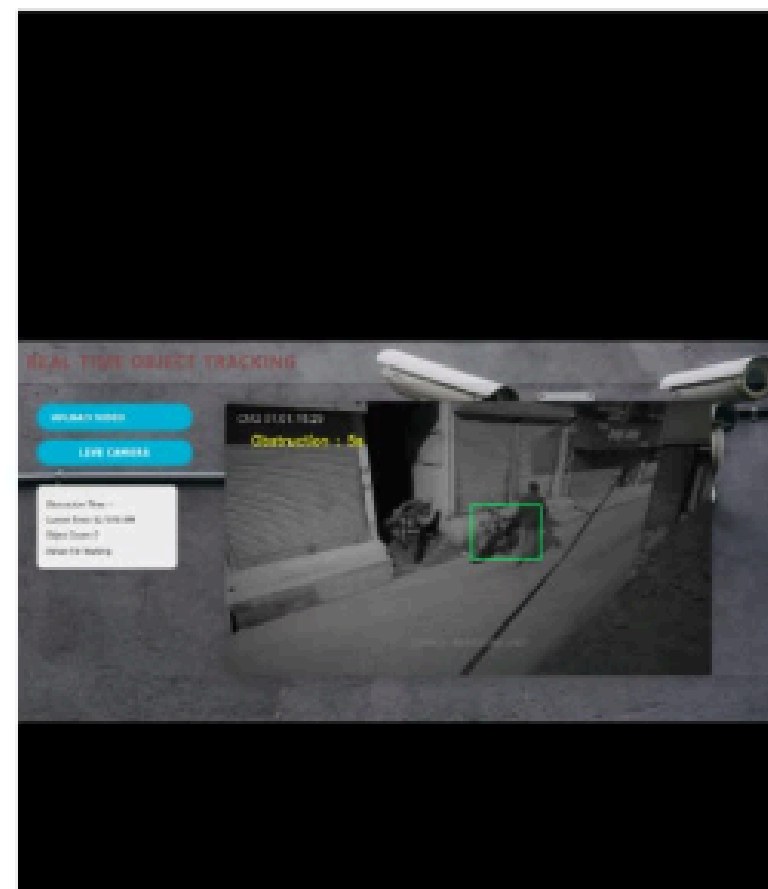
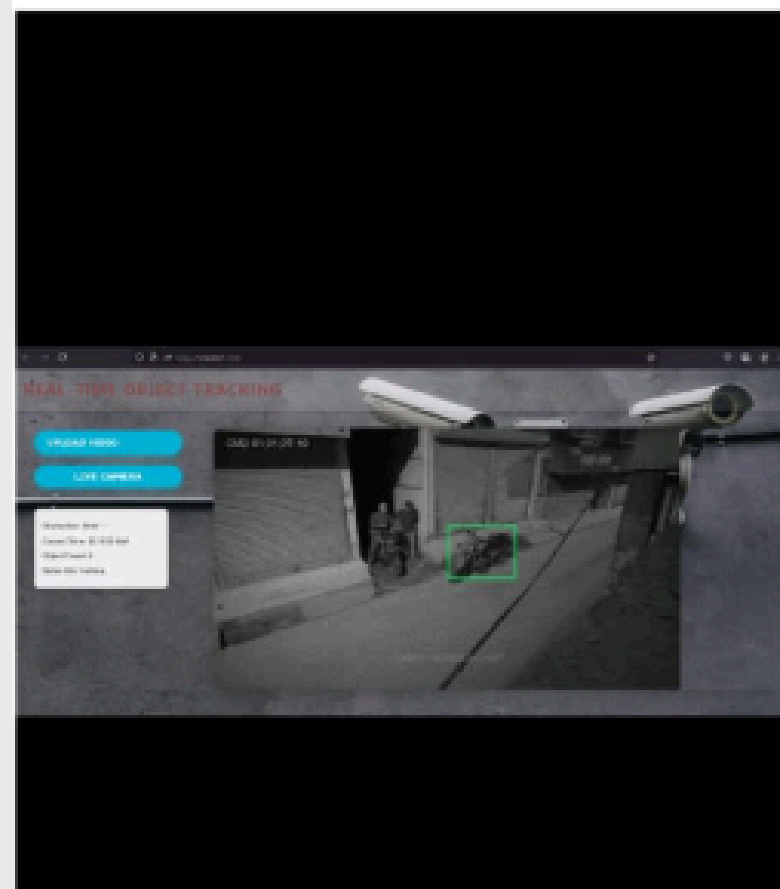
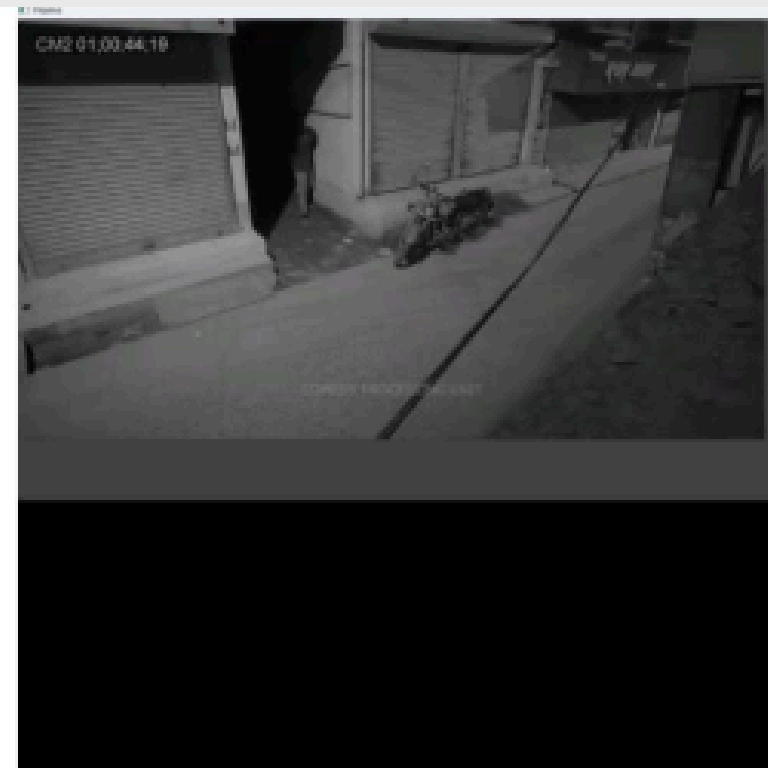
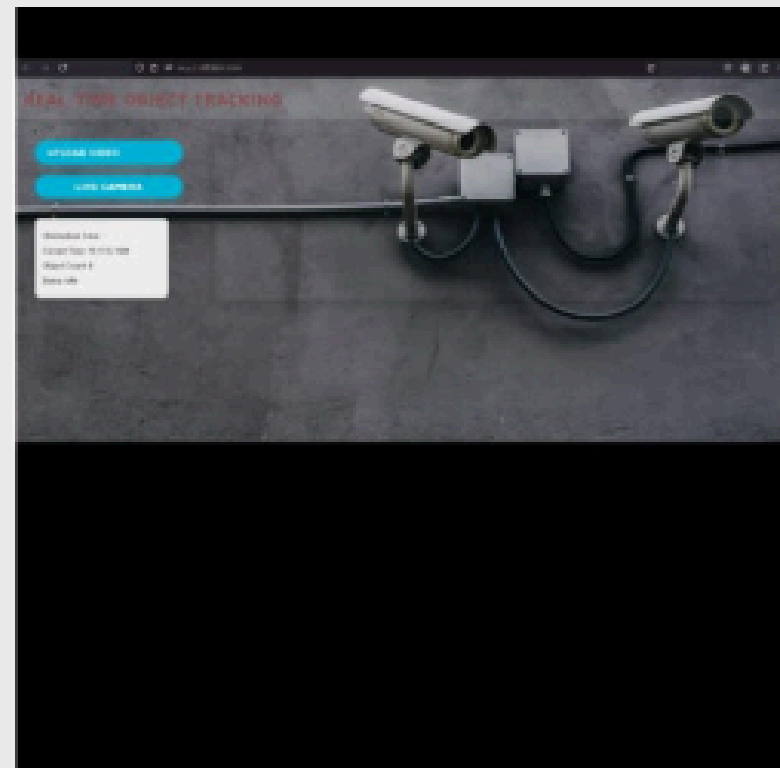
HTML, CSS, JavaScript, Fetch API.

3

Algorithms

MOSSE, KCF, CSRT, Frame differencing.

RESULTS



conclusion

The proposed AI-powered security surveillance system offers an innovative and effective solution to the growing challenges of theft and unauthorized movement of vehicles and valuable assets. By integrating advanced deep learning models with infrared and thermal imaging, the system ensures reliable performance in low-light and visually obstructed environments. Real-time detection and alert capabilities enable faster response times and proactive threat mitigation. This intelligent surveillance approach not only enhances the safety and security of personal and commercial properties but also sets the foundation for future smart security systems that are adaptive, efficient, and highly responsive.

Let me know if you'd like to adjust this for a presentation or re

New Features: Enhanced Monitoring



Real-Time Video

MJPEG streaming for live video feed.



Movement Detection

Alerts when object moves significantly.



Obstruction Detection

Tracks duration and records initial time.

REFERENCES

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